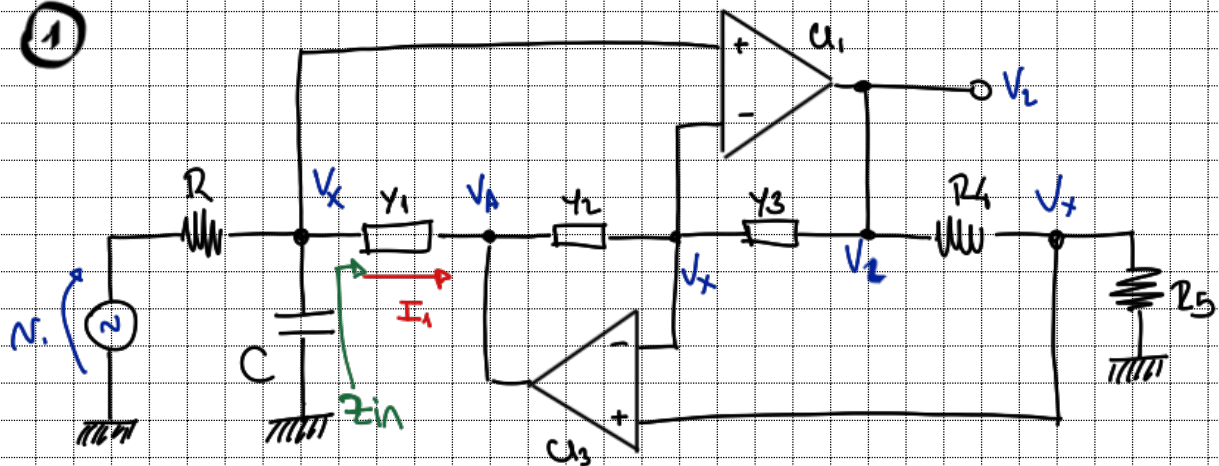


Tarea 1

Montenegro Maximiliano

1



$$\begin{cases} (V_x - V_A) Y_1 = I_1 \\ (V_A - V_x) Y_2 = (V_x - V_2) Y_3 \\ (V_2 - V_x) Y_4 = V_x Y_5 \end{cases}$$

$$(V_2 - V_x) Y_4 = V_x Y_5$$

$$V_x (Y_4 + Y_5) = V_2 Y_4$$

$$V_2 = V_x R_4 (Y_4 + Y_5)$$

$$(V_A - V_x) Y_2 = (V_x - V_2) Y_3$$

$$V_A Y_2 = V_x (Y_2 + Y_3) - V_2 Y_3$$

$$V_A Y_2 = V_x (Y_2 + Y_3) - V_x Y_3 R_4 (Y_4 + Y_5)$$

$$V_A Y_2 = V_x (Y_2 + Y_3 - Y_3 R_4 (Y_4 + Y_5))$$

$$V_A Y_2 = V_x (Y_2 + Y_3 - Y_3 R_4 Y_4 - Y_3 R_4 Y_5)$$

$$V_A Y_2 = V_x (Y_2 + Y_3 - Y_3 R_4 Y_5)$$

$$V_A = Z_2 V_x (Y_2 - Y_3 R_4 Y_5)$$

$$V_A = V_x (Z_2 Y_2 - Z_2 Y_3 R_4 Y_5)$$

$$V_A = V_x (1 - Z_2 Y_3 R_4 Y_5)$$

$$(V_x - V_A) Y_1 = I_1$$

$$V_x - V_x (1 - Z_2 Y_3 R_4 Y_5) = G_1 I_1$$

$$V_x (1 - 1 + Z_2 Y_3 R_4 Y_5) = G_1 I_1$$

$$; Z_{in} = \frac{V_x}{I_1}$$

$$\frac{V_x}{I_1} = Z_{in} = \frac{Z_2 Y_3 R_4}{Z_2 Y_4}$$

$$Z_{in} = \frac{Z_1 Z_3 R_5}{Z_2 R_4}$$

$$Z_R(s) = R$$

$$Z_L(s) = sL$$

$$Z_C(s) = \frac{1}{sC}$$

para que se comporte

como un inductor

debe tener un cero $\therefore$

$$Z_2 = \frac{1}{sC_2} ; \quad Z_1 = Z_3 = R$$

$$Z_{in_L} = \frac{R_1 R_3 R_5}{\frac{1}{sC_2} R_4}$$

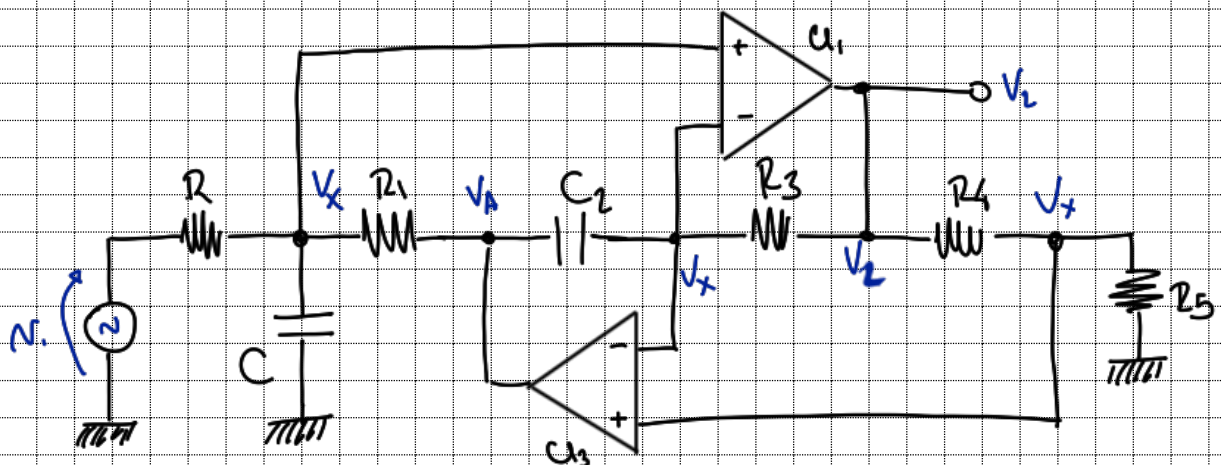
$$Z_{in_L} = sC_2 \frac{R_1 R_3 R_5}{R_4}$$

$$L = \frac{C_2 R_1 R_3 R_5}{R_4} ;$$

$$L = C_2 \frac{R_1 R_3 R_5}{R_4}, \text{ si deseo } L=1$$

$$\Rightarrow 1 = C_2 \frac{R_1 R_3 R_5}{R_4} : \quad C_2 = \frac{R_4}{R_1 R_3 R_5}$$

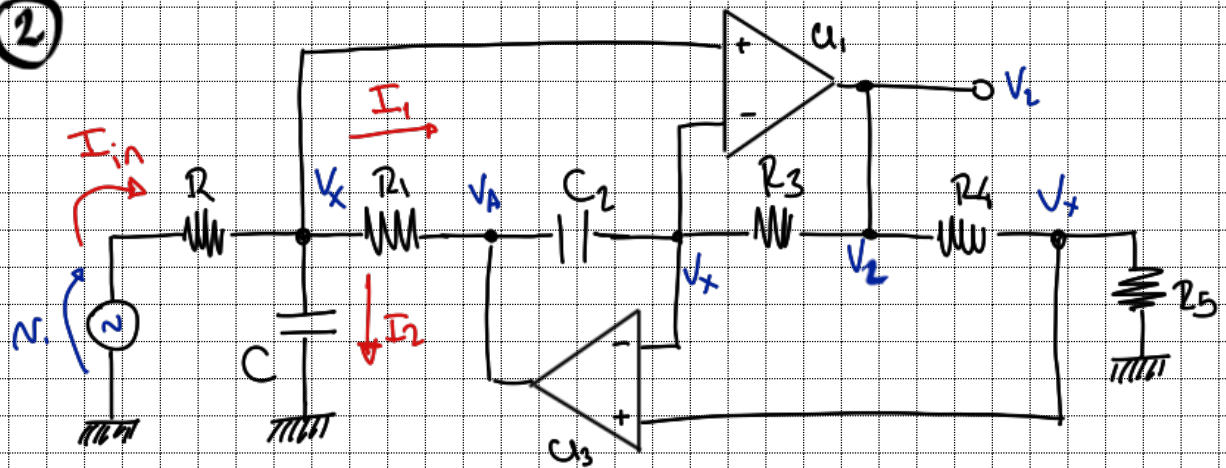
Quedando:



$$R_1 = R_4 = R_5 = 1 \text{ k}\Omega ; \quad R_3 = 10 \text{ k}\Omega$$

$$C_2 = 100 \text{ nF}$$

②



$$\begin{cases} V_x (G + G_1 + SC) = V_1 G + V_A G_1 & (1) & V_A = R_1 (V_x (G + G_1 + SC) - V_1 G) \\ V_x (SC_2 + G_3) = V_A SC_2 + V_2 G_3 & (2) & V_A = V_x (R_1 G + R_1 G_1 + SC_2 C) - V_1 R_1 G \\ V_x (G_4 + G_5) = V_2 G_4 & (3) & V_A = V_x (1 + R_4 G + SC R_1 C) - V_1 R_1 G \end{cases}$$

$$\begin{aligned} (1) \quad V_x (SC_2 + G_3) &= (V_x (1 + R_1 (G + SC)) - V_1 R_1 G) SC_2 + V_2 G_3 \\ V_x (SC_2 + G_3) &= V_x SC_2 (1 + R_1 (G + SC)) - V_1 R_1 G SC_2 + V_2 G_3 \\ V_x (SC_2 + G_3 - SC_2 - SC R_1 C_2 (G + SC)) &= -V_1 R_1 G SC_2 + V_2 G_3 \\ V_x (G_3 - SC R_1 C_2 (G + SC)) &= -V_1 SC R_1 G C_2 + V_2 G_3 \\ V_x &= \frac{V_2 G_3 - V_1 SC R_1 G C_2}{(G_3 - SC R_1 C_2 (G + SC))} \end{aligned}$$

$$\begin{aligned} (3) \quad \frac{V_2 G_3 - V_1 SC R_1 G C_2}{G_3 - SC R_1 C_2 (G + SC)} (G_4 + G_5) &= V_2 G_4 \\ \frac{V_2 G_3 (G_4 + G_5)}{G_3 - SC R_1 C_2 (G + SC)} - V_2 G_4 &= \frac{V_1 SC R_1 G C_2 (G_4 + G_5)}{G_3 - SC R_1 C_2 (G + SC)} \\ V_2 (G_3 (G_4 + G_5) - G_4 (G_3 - SC R_1 C_2 (G + SC))) &= \frac{V_1 SC R_1 C_2 (G_4 + G_5) G}{G_3 - SC R_1 C_2 (G + SC)} \\ V_2 (G_3 G_4 + G_3 G_5 - G_3 G_4 + SC R_1 G_4 C_2 (G + SC)) &= V_1 SC R_1 C_2 (G_4 + G_5) G \\ \frac{V_2}{V_1} &= A_v = \frac{SC R_1 C_2 (G_4 + G_5) G}{G_3 G_5 + SC R_1 G_4 C_2 (G + SC)} \\ A_v &= \frac{SC R_1 C_2 (G_4 + G_5) G}{S^2 R_1 G_4 C_2 C + SC R_1 G_4 C_2 G + G_3 G_5} \end{aligned}$$

$$A_V = \frac{S R_1 C_2 (G_4 + G_5) G}{S^2 R_1 G_4 C_2 C + S R_1 G_4 C_2 G + G_3 G_5}$$

$$A_V = \frac{S R_1 C_2 \left(\frac{1}{R_4} + \frac{1}{R_5} \right) G}{S^2 \frac{R_1}{R_4} C_2 C + S \frac{R_1}{R_4} C_2 \frac{1}{R} + \frac{1}{R_3} \frac{1}{R_5}}$$

$$\frac{S R_1 C_2 \left(\frac{R_4 + R_5}{R_4 R_5} \right) G}{S^2 \frac{R_1}{R_4} C_2 C + S \frac{R_1}{R_4} C_2 \frac{1}{R} + \frac{1}{R_3} \frac{1}{R_5}}$$

$$\frac{\frac{1}{R_4 R_5} \cdot S R_1 C_2 (R_4 + R_5) G}{\frac{1}{R_3 R_4 R_5} \left(S^2 R_1 R_3 R_5 C C_2 + S R_1 R_3 R_5 C_2 + R R_4 \right)}$$

$$A_V = \frac{S R_1 R_3 C_2 (R_4 + R_5)}{S^2 R_1 R_3 R_5 C C_2 + S R_1 R_3 R_5 C_2 + R R_4}$$

$$A_V = \frac{\cancel{R_1} \cancel{R_3} C_2 (R_4 + R_5)}{\cancel{R_1} \cancel{R_3} \cancel{R_5} C C_2} \cdot \frac{S}{\left(S^2 + S \frac{1}{R C} + \frac{R_4}{R_1 R_3 R_5 C C_2} \right)}$$

$$A_V = \frac{R_4 + R_5}{R R_5 C} \cdot \frac{S}{S^2 + S \frac{1}{R C} + \frac{R_4}{R_1 R_3 R_5 C_2 C}}$$

$$\frac{\omega_0}{Q} = \frac{1}{R C} =$$

$$\omega_0^2 = \frac{R_4}{R_1 R_3 R_5 C_2 C}$$

$$H(s) = \frac{R_4 + R_5}{R R_5 C} \frac{S}{S^2 + S \frac{\omega_0}{Q} + \omega_0^2}$$

$$H(s=j\omega) = \frac{R_4 + R_5}{R R_5 C} \frac{j\omega}{-j\omega + j\omega \frac{\omega_0}{Q} + \omega_0^2}$$

$$H(j\omega) = \frac{R_4 + R_5}{R R_5 C} \frac{j\omega}{\omega_0^2 - \omega^2 + j\omega \frac{\omega_0}{Q}}$$

$$|H(j\omega)| = \frac{R_4 + R_5}{R R_5 C} \frac{\omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + \left(\frac{\omega \omega_0}{Q}\right)^2}}$$

$$|H(\omega_0)| = \frac{R_4 + R_5}{R R_5 C} \frac{\omega_0}{\sqrt{(\omega_0^2 - \omega_0^2)^2 + \left(\frac{\omega_0 \omega_0}{Q}\right)^2}}$$

$$\frac{R_4 + R_5}{R R_5 C} \frac{\omega_0}{\left(\frac{\omega_0^2}{Q}\right)^2}$$

$$\frac{R_4 + R_5}{R R_5 C} \frac{\omega_0}{\frac{\omega_0^2}{Q}}$$

$$\frac{R_4 + R_5}{R R_5 C} \frac{Q}{\omega_0} \quad RC$$

$$|H(\omega_0)| = \frac{R_4 + R_5}{R R_5 C} \frac{Q}{\omega_0}$$

$$20 \text{ dB} = 20 \log(|H(\omega_0)|)$$

$$10 \left(\frac{20 \text{ dB}}{20} = \log\left(\frac{R_4 + R_5}{R_5}\right) \right)$$

$$10 = \frac{R_4 + R_5}{R_5}$$

$$10 R_5 = R_4 + R_5$$

$$9 R_5 = R_4$$

$$\frac{\omega_0}{Q} = \frac{1}{RC}$$

$$R = \frac{Q}{2\pi f_0 C} \quad C = 47 \text{ nF}$$

$$R = \frac{20}{2\pi \cdot 10 \cdot 10^3 \frac{1}{s} \cdot 47 \cdot 10^{-9} \frac{C}{V}} = 6772,5 \, \Omega = R$$

$$\omega_0^2 = \frac{R_4}{R_1 R_3 R_5 C_2 C}$$

$$R_4 = 9 \text{ k}\Omega$$

$$C_2 = 1 \text{ nF}$$

$$R_1 R_3 = \frac{9 R_5}{R_5 (2\pi f_0)^2 C_2 C}$$

$$R_1 R_3 = \frac{9}{(2\pi \cdot 10 \cdot 10^3 \frac{1}{s})^2 \cdot 1 \cdot 10^{-9} \text{ F} \cdot 47 \cdot 10^{-9} \text{ F}} \approx 10504821,96 \Omega^2$$

$$R_1 = R_3 \approx 6964,54 \Omega$$

$$R_4 = 9 \text{ k}\Omega$$

$$R_5 = 1 \text{ k}\Omega$$

$$R = 6772,5 \Omega$$

$$C = 47 \text{ nF}$$

$$C_2 = 1 \text{ nF}$$